DSAI 5810/6810 - Intro to NLP

Fall Semester 2025

Instructor

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- Contact: a00980125@usu.edu
- Office Hours: Wednesday at 8 on Zoom. Please email me at least by day of if you plan on coming.

Prerequisites

Instructor permission in some cases.

Course Materials

No textbook needed for this course. Supplementary materials will be provided as necessary.

Introduction

Artificial Intelligence (AI) is one of the most transformative technologies of the past few decades. While the extent and nature of its impact on society remains a subject of debate, it is clear that AI is here to stay—and its influence will likely grow even stronger in the years ahead. It is therefore imperative for students, professionals and others to more fully understand AI, its origins and how to use it.

One of the main founding fields and driver of AI advancements is Natural Language Processing (NLP). NLP focuses on enabling machines to understand, interpret, and generate human language, making it essential for anyone seeking to understand modern AI.

After finishing this course, students should be able to:

- Understand basic NLP techniques for analyzing and generating text.
- Understand how to build classical NLP models and how those concepts led to current AI.
- Understand the basic mathematical underpinnings of AI (neural nets, the transformer, etc.)
- Understand how to build a basic, modern AI application.

This is an *Intro* (note the emphasis) course. I hope to introduce you to as many tools and ideas as possible, knowing that a fuller understanding of concepts will only come through practice and application in future courses, research, work settings, and/or personal interest. I have tried to weigh the course content towards focusing on the most current and relevant topics in industry, while still providing a background on traditional techniques (which still have value in industry today) and how they have led to current approaches.

With the above in mind, the focus of this course will be more on *application* vs *theory*. Both are important, and both will be taught, but focusing on application is where the "rubber meets the road" and where better understanding can occur. The other hope with this approach is to provide you with basic experience (application), balanced with basic knowledge (theory) that will make you more competitive in the job market.

Software

This course will almost exclusively be in Python. All students will be required to have a working laptop capable of installing Python that they can bring to each class period. Other modern dev tools will be required such as Git. There will also be services students will need to create an account for including Github, Google AI, etc. I will help introduce and set up these tools during the course.

Grades

Final scores in this course will be determined by the number of points earned divided by 1000. Thus, each 10 points on an assignment, regardless of category, represents 1% of your final score. The following grade scale is guaranteed, but I reserve the right to adjust this grade scale in favor of students if necessary. Note that this grade scale contains no + or - designations. This means that a 91% is an A, not an A- and an 89% is a B, not a B+, in this course.

Percent	Grade
90-100	A
80-89.9	B
70-79.9	C
60-60.9	D
0-59.9	F

Assignments

The general expectation for a one credit class is about 2-3 hours of work outside of class per week. I fully intend to keep the workload of this class at or below that level. If you find the assignments are taking longer than 2-3 hours, please let me know sooner than later so adjustments can be made.

With this in mind I've split assignments into three sections:

In-class Homework (750 points)

I'm a big believer in learning by doing, and so I aim to code and work together with you in class on assignments. These assignments will be due on the day of the following class period, and may require completion outside of class, but if you come to class you should complete most of the assignment or at least have a good direction. If you have to miss class, you will be expected to complete and submit the in-class assignments on your own outside of class.

Points will be given for well-organized, documented, and well-formatted code that runs correctly.

I'm aiming for 10 assignments, so not every class will have an associated assignment. I will drop your lowest score.

AI Application RAG Project (200 points)

This project will require you to build a basic RAG (Retrieval-Augmented Generation) question/answering application. The project will help you piece together different topics we'll learn over the course and also provide "real-world" experience when it comes to gathering data, cleaning, processing, piecing together APIs, prompts, etc. It will be due at the end of the semester, but we will start mid-semester to help make the project more digestible.

Hot-Seat Questions (50 points)

Two important components of learning and retaining (of many) include 1) reviewing previous material and 2) teaching the material to others. Also, being able to verbally and succinctly describe a concept to others is essential for job interviews, presentations, and generally impressing friends, family or your next date. To that end, at the beginning of each class I'll randomly choose a few of you to verbally answer questions posed at the end of the last class period. This should be considered a low-stress exercise - as long as I can tell you made an effort to think about the question you'll get full points.

Late Policy

All assignments are due by 11:59 pm MDT on the listed due date in Canvas. No late assignments will be accepted, but I also know "life happens" so please reach out sooner than later for any extenuating circumstances, such as medical, personal, family emergencies, etc.

Instructor Commitments

I have a full-time industry job, along with a personal/family life, so I won't always be able to provide immediate responses to emails, etc. I will do my best however to respond within one business day, but may not respond to your requests over the weekend.

Student Expectations

Attendance

I once heard someone say "education is a product that we pay for, but don't necessarily want the full value of". There is no official attendance policy - it is your time and money. I do believe however, almost every subject discussed in class should be directly relevant to the "real world" and to help prepare you for any future path in NLP/AI. That's on me to make that true and to make the class generally engaging, but it's also on you to engage and get out of the class what you can. In short, attendance does matter.

Course Communication

Students are responsible for all information presented in class and are responsible to regularly check Canvas for announcements, due dates, and other communications from the instructor.

Asking Questions

Students are strongly encouraged to ask questions during class and office hours. Students are also encouraged to ask questions related to course assignments using the “Discussions” tab on the course Canvas page. Posting assignment-related questions on the discussion board provides opportunities for other members of the class to answer questions. Keep in mind, however, that you should NOT post entire blocks of code when answering or asking questions. The discussion boards work best when all of us actively engage in both asking questions and providing answers. Note that personal/sensitive questions should not be posted to the discussion board but rather sent as a Canvas email message.

Academic Honesty

In this course, cheating is defined as any attempt made by a student to deceive the instructor in the representation of their work. This definition includes all forms of plagiarism, including copying code from other students in the course. It is natural and encouraged to take programming ideas and small code snippets from online sources or AI, but please refrain from having AI write your entire assignment. Any observed instances of cheating will be reported to the University and warrants automatic failure from the course.

Recommendations for Course Success

It is critical to remember that understanding the concepts discussed in this course will come through repetition. Rarely, if ever, is a complex concept understood after hearing it once in lecture. Rather, understanding will come by:

- Running the provided programming examples on your own machine.
- Engaging in class discussions by asking and answering questions.
- Reviewing material between class sessions.
- Starting homework assignments soon after they are assigned.

IDEA Objectives

The university using the IDEA learning objectives to measure the effectiveness of courses. Of the 13 pre-defined learning objectives, the ones most relevant to this course include:

- (1) Gaining a basic understanding of the subject (e.g., factual knowledge, methods, principles, generalizations, theories)
- (3) Learning to apply course material (to improve thinking, problem solving, and decisions)
- (13) Learning appropriate methods for collecting, analyzing, and interpreting numerical information

Disclaimer

This is a new course that may require mid-semester adjustments. As such, the instructor reserves the right to make minor changes to this syllabus as needed throughout the semester. No change in the syllabus will be made without first advertising the proposed change to the students. Students will be given 2-3 days to review any changes and provide comments/objections to the instructor before any change is executed.

University Resources

Disability Resource Center

USU Welcomes students with disabilities. If students have, or suspect they may have, a physical, mental health, or learning disability that may require accommodations in this course, please contact the Disability Resource Center (DRC) as early in the semester as possible (University Inn #101, 435-797-2444, drc@usu.edu, \url{usu.edu/drc}). All disability-related accommodations must be approved by the DRC. Once approved, the DRC will coordinate with faculty to provide accommodations.

Office of Equity

USU strives to provide an environment for students and employees that is free from discrimination or harassment, including sexual misconduct. Should students experience harassment or discrimination at any point during the semester inside or outside of class, please reach out to the instructor or to USU's Office of Equity (Old Main #161, 435-797-1266, titleix@usu.edu, \url{equity.usu.edu}). As a responsible employee, the instructor is required to share information about any instances of sexual misconduct (sexual harassment, sexual assault, relationship violence (dating and domestic violence, or stalking) with the Office of Equity so that students can get connected to support and reporting resources. Students can learn more about the USU resources available for individuals who have experienced sexual misconduct at \url{sexualassault.usu.edu}.

CAPS

Mental health is critically important for the success of USU students. As a student, you may experience a range of issues that can cause barriers to learning, such as strained relationships, increased anxiety, alcohol/drug problems, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may lead to diminished academic performance or reduce your ability to participate in daily activities. Utah State University provides free services for students to assist them with addressing these and other concerns. You can learn more about the broad range of confidential mental health services available on campus at Counseling and Psychological Services (CAPS).

Students are also encouraged to download the "SafeUT App" to their smartphones. The SafeUT application is a 24/7 statewide crisis text and tip service that provides real-time crisis intervention to students through texting and a confidential tip program that can help anyone with emotional crises, bullying, relationship problems, mental health, or suicide related issues.